

Is this Idea Novel? An Automated Benchmark for Judgment of Research Ideas

Tim Schopf^{1,2} and Michael Färber¹

¹TU Dresden & ScaDS.AI Dresden/Leipzig, Germany

²National Institute of Informatics, Tokyo, Japan
{tim.schopf, michael.färber}@tu-dresden.de

Abstract

Judging the novelty of research ideas is crucial for advancing science, enabling the identification of unexplored directions, and ensuring contributions meaningfully extend existing knowledge rather than reiterate minor variations. However, given the exponential growth of scientific literature, manually judging the novelty of research ideas through literature reviews is labor-intensive, subjective, and infeasible at scale. Therefore, recent efforts have proposed automated approaches for research idea novelty judgment. Yet, evaluation of these approaches remains largely inconsistent and is typically based on non-standardized human evaluations, hindering large-scale, comparable evaluations. To address this, we introduce RINoBench, the first comprehensive benchmark for large-scale evaluation of research idea novelty judgments. It comprises 1,381 research ideas derived from and judged by human experts as well as nine automated evaluation metrics designed to assess both rubric-based novelty scores and textual justifications of novelty judgments. Using this benchmark, we evaluate several state-of-the-art large language models (LLMs) on their ability to judge the novelty of research ideas. Our findings reveal that while LLM-generated reasoning closely mirrors human rationales, this alignment does not reliably translate into accurate novelty judgments, which diverge significantly from human gold standard judgments—even among leading reasoning-capable models. Data and code available at: <https://github.com/TimSchopf/RINoBench>.

Keywords: research idea novelty judgment, evaluation benchmark, scientific discovery, llm-as-a-judge

1. Introduction

Judging the novelty of research ideas is fundamental to fostering scientific discovery and ensuring that new works meaningfully advance a field rather than reproducing existing results with minor variations that contribute little new insight. Hence, effective novelty judgment helps researchers identify unexplored directions, develop original contributions, and ultimately drive scientific progress. However, manually judging the novelty of a research idea requires a comprehensive review of previous work to determine whether the same or similar ideas have already been explored. With the rapid growth of scientific literature (Fortunato et al., 2018), this manual process has become increasingly challenging for researchers in terms of both time and cognitive effort. Moreover, novelty judgments are inherently subjective. Experts can often identify when two ideas are similar (Picard et al., 2025) but struggle to articulate what makes an idea truly novel (Shahid et al., 2025). In addition, such judgments are influenced by an individual’s prior knowledge, intuition, and familiarity with the relevant literature (Ahmed et al., 2018; Picard et al., 2025). To address these challenges, automated approaches have been proposed to support and enhance research idea novelty judgments.

Recent work has used large language models (LLMs) to automatically judge the novelty of research ideas (Lu et al., 2024; Si et al., 2025; Li

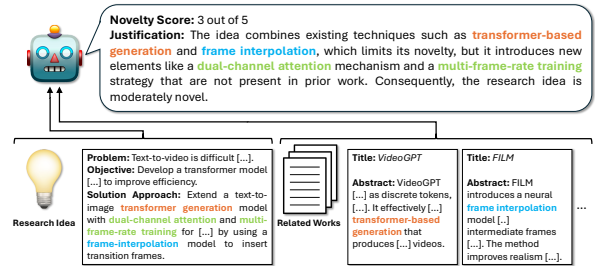


Figure 1: The task setup of RINoBench. Given a research idea and its related works, a model must judge the novelty of the idea according to a five-point rubric. In addition, the model must provide a textual justification for its judgment, grounded in a comparison between the proposed research idea and the related works.

et al., 2025a; Su et al., 2025; Gottweis et al., 2025). However, these approaches do not ground their rationales in prior literature and struggle with subtle linguistic variation, leading to the misclassification of well-established ideas as novel (Beel et al., 2025; Gupta and Pruthi, 2025; Wang et al., 2025b). Furthermore, many LLM-based approaches restrict their outputs to binary classifications (novel vs. not novel) (Lu et al., 2024; Li et al., 2025a; Shahid et al., 2025; Su et al., 2025), overlooking the nuanced and gradual nature of novelty judgments. Moreover, most automated approaches provide only final pre-

dictions without offering interpretable explanations or justifications supporting their decisions. This lack of transparency reduces their practical utility, as researchers cannot review or learn from opaque judgments, hindering their ability to refine research ideas toward greater novelty. Finally, the fundamental limitations and differences between existing automated research idea judgment approaches make meaningful comparisons difficult. This problem is exacerbated by the fact that current evaluations of automated research idea judgment approaches are mainly based on non-standardized manual evaluations (Si et al., 2025; Gottweis et al., 2025, *inter alia*), hindering large-scale, systematic comparisons.

To address these limitations, we introduce the **Research Idea Novelty Judgment Benchmark** (RINoBench¹), the first *comprehensive* and *reproducible* benchmark for the automatic evaluation of research idea novelty judgments. Using this benchmark, we conduct the first large-scale benchmarking study of current state-of-the-art LLMs, evaluating their ability to judge the novelty of research ideas. We reveal that *while LLMs often generate reasoning patterns similar to human experts, they fail to consistently translate these rationales into accurate novelty judgments*.

Our main contributions are:

- RINoBench, a comprehensive benchmark for systematically evaluating research idea novelty judgments, comprising **1,381 research ideas derived from and judged by human experts** as well as **nine automated evaluation metrics** designed to assess both rubric-based novelty scores and textual justifications of novelty judgments.
- A study investigating several state-of-the-art LLMs on their ability to judge the novelty of research ideas, involving a systematic analysis of the strengths and limitations of current state-of-the-art LLMs performing this task.

2. Related Work

Automated methods for judging novelty in scientific literature have advanced significantly in recent years. Early approaches measured novelty via atypical combinations of cited references (Uzzi et al., 2013), constructed historical co-occurrence matrices and derived journal vectors, where lower cosine similarity indicates greater novelty (Wang et al., 2017), and relied on lexical similarity (Wang et al., 2019; Sarica et al., 2020). However, such approaches are inherently limited in their ability to capture paraphrased, conceptually equivalent, or closely related ideas, as they reduce novelty to patterns of statistical co-occurrence or lexical overlap rather than accounting for their semantic

relationships. Subsequent work using semantic embeddings (Gómez-Pérez et al., 2022) enhanced the ability to identify deeper conceptual relations, but remains constrained to surface-level semantic comparisons (Mysore et al., 2022). More recently, Wu et al. (2025b) combined human and LLM knowledge for novelty evaluation. In addition, retrieval-augmented LLM approaches have emerged as promising alternatives (Yang et al., 2024; Bougie and Watanabe, 2024; Lu et al., 2024; Radensky et al., 2024; Si et al., 2025; Liu et al., 2025; Su et al., 2025; Wang et al., 2025a; Baek et al., 2025; Zhang et al., 2025; Tang et al., 2025; Li et al., 2025a, *inter alia*). However, these approaches typically treat novelty judgment of research ideas as an intermediate step within a broader AI-assisted scientific discovery pipeline. As a result, and further exacerbated by the lack of automated benchmarks, these works either omit a dedicated and systematic evaluation of their novelty judgments or rely on costly and often small-scale human evaluations. Complementary to this, Wen et al. (2025) introduce a large-scale benchmark dataset designed to predict which of two research ideas performs better on a given set of benchmarks, but they do not address the task of novelty judgment. In addition, Wu et al. (2025a) examine which sections of research papers are most informative for novelty judgment. The work most closely related to ours provides the only publicly available evaluation dataset to date for judging the novelty of research ideas (Shahid et al., 2025). However, this dataset is limited to 51 manually annotated research ideas with binary labels (novel vs. not novel) and does not include any evaluation of textual novelty judgment justifications.

3. On the Notion of Novelty

Novelty is a fundamental concept in scientific research, which has been extensively characterized in existing literature. Arts et al. (2021) consider novelty as the uniqueness of specific knowledge elements, whereby the introduction of previously unknown elements indicates novel information. Further, Foster et al. (2021) define novelty as the extent to which a proposed contribution diverges from the existing scientific literature. Importantly, novelty is not limited to entirely new knowledge. An idea can also be considered novel if it represents a previously unobserved combination of known knowledge elements or applies them to new contexts (Boudreau et al., 2016; Shahid et al., 2025).

Closely related to novelty is the concept of *originality*. It refers to the generation of new ideas, methods, conclusions, or other valuable outputs that deviate from existing paradigms and can inspire further innovation (Shibayama and Wang, 2019; Hou et al., 2022). In practice, however, distinguish-

ing originality from novelty is challenging (Guetzkow et al., 2004), leading to the frequent interchangeable use of these terms (Wu et al., 2025a).

In essence, novelty, often used interchangeably with originality, is a fundamental driver of scientific progress, providing the foundation for both innovation and disruptive advances. While a novel idea frequently entails introducing a previously unseen element of knowledge, it can also emerge from a previously unexplored combination of existing knowledge in innovative ways.

4. Benchmark

RINoBench unifies approaches for judging the novelty of research ideas by formalizing the task, illustrated in Figure 1, as the process of comparing a proposed research idea with existing work to identify meaningful differences. Further, the task requires predicting a rubric-based novelty score (1–5) alongside a textual justification that grounds the judgment in related literature.

4.1. Data

Collecting a comprehensive dataset through dedicated workshops or user studies is prohibitively expensive and practically infeasible due to the complexity of the task. Human experts would need to generate novel research ideas, and other experts would then need to evaluate them. Both tasks impose a high cognitive load and require substantial domain expertise, meaning that each instance demands significant time and effort. Moreover, the pool of qualified participants is small, making recruitment difficult. Consequently, prior data collection efforts of this kind have been limited in scale, typically yielding only around 50 human-generated research ideas (Si et al., 2025; Shahid et al., 2025).

To overcome these limitations, we adopt a different strategy by leveraging publicly available data from OpenReview. Specifically, peer reviews from ICLR 2022 and ICLR 2023 provide a rich source of information: human experts have already submitted papers based on their research ideas, which have been explicitly evaluated by other human experts using rubric-based novelty scores and corresponding textual justifications. By processing and enriching this peer review data, we construct a high-quality dataset for studying research idea novelty judgment.

4.1.1. Data Collection & Processing

We collect all publicly available ICLR 2022 and ICLR 2023 submissions and their corresponding reviews from OpenReview, yielding 6,410 papers with associated reviewer feedback. Each submission was evaluated by approximately three expert

reviewers, who rated the novelty of the research using a rubric-based numerical scale and provided brief textual justifications. Specifically, reviewers assessed two novelty dimensions: “*Technical Novelty and Significance*” and “*Empirical Novelty and Significance*”. We use both dimensions in our dataset. Since human novelty judgments are inherently subjective and may vary substantially, we filter out all submissions where the maximum disagreement between reviewers exceeds one point within and across both novelty dimensions. This results in a filtered dataset of 3,535 paper submissions with high inter-reviewer agreement.

To obtain a single gold-standard novelty score for each paper, we average all reviewer scores across both novelty dimensions. The resulting decimal values, however, are difficult to interpret and predict. To address this, we transform them into whole numbers on a unified 1–5 scoring rubric by binning the averaged values into five intervals. This 1-5 scoring rubric, as shown in Table 1, offers an intuitive and standardized measure of novelty, featuring a clear midpoint, balanced polarity, and nuanced gradation, consistent with conventions commonly used in user studies.

Score	Degree of Novelty
1	The idea is not novel. All aspects already exist in prior work.
2	The idea is marginally novel. It represents only a minor variation of existing work.
3	The idea is somewhat novel. Aspects already exist in prior work. However, it might combine known approaches in new ways, apply them to new contexts, or propose incremental updates.
4	The idea is novel. It introduces new aspects not present in existing work.
5	The idea is highly innovative and novel. It is not present in existing work and potentially encourages new thinking or opens up new research directions.

Table 1: Novelty Judgment Rubric

Our next data processing step focuses on transforming submitted papers into concise research ideas by systematically identifying and reformulating their core ideas and contributions. Specifically, we provide an LLM with paper titles, abstracts, and reviewer summaries as context to distill the most salient information and produce structured and concise representations of the underlying research ideas. For this and all subsequent LLM-based data processing steps, we use the *GPT-OSS-120B* (OpenAI et al., 2025) model. In this step, the model is prompted to analyze the provided context, identify the key elements that define a paper’s research idea, and output a structured JSON representation